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Can a Large Language Model Interpret Data in the Electronic Health Record to Infer Minimum Clinically Important Difference Achievement of Knee Osteoarthritis Outcome Score–Joint Replacement Score Following Total Knee Arthroplasty?



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ABSTRACT

Background: Obtaining total knee arthroplasty patient-reported outcomes for quality assessment is costly and difficult. We asked whether a large language model (LLM) could interpret electronic health record notes to differentiate patients attaining a 1-year minimum clinically important difference (MCID) for the Knee Osteoarthritis Outcome Score–Joint Replacement (KOOS-JR) from those who did not. We also investigated whether sufficient information to infer MCID achievement exists in the chart by having a blinded orthopaedic surgeon make the same determination.

Methods: In this retrospective case-control study, we selected 40 total knee arthroplasty patients who achieved 1-year KOOS-JR MCID and 40 who did not. Orthopaedic, emergency medicine, and primary care notes from zero to six months preoperatively and nine to 15 months postoperatively were deidentified. ChatGPT 3.5 (ChatGPT) interpreted these notes to determine whether the patient improved after surgery. A blinded orthopaedic surgeon classified these patients using all chart information. The sensitivity, specificity, and accuracy of ChatGPT and the surgeon's responses were calculated.

Results: ChatGPT classified 78 of 80 cases with 97% sensitivity, but only 33% specificity. The surgeon's assessment had 90% sensitivity and 63% specificity. Given the equal distribution of patients meeting or not meeting MCID, Chat GPT's accuracy was 65%. The surgeon's was 76%.

Conclusions: ChatGPT's assessment of KOOS-JR MCID attainment had 97% sensitivity, but only 33% specificity. False positives were commonly due to the LLM not having access to, or not properly interpreting, signs of problems in the chart. This was an initial evaluation of the current ability of a general-purpose LLM to evaluate patient outcomes based on information in chart notes. An orthopaedic surgeon's assessment of the full chart suggests an opportunity to improve on this baseline performance, possibly enabling quality monitoring and identification of best practices across a large health care system. Additional work is needed to optimize model performance and confirm the utility of this approach.

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Total knee arthroplasty (TKA) is a well-established treatment for end-stage osteoarthritis, but not all patients improve after surgery. The efficacy of TKA is commonly assessed through

patient-reported outcome measures (PROMs), which reflect the patient's subjective evaluation of their postsurgical recovery and quality of life [1,2]. Although PROMs are considered the gold standard measure of patient outcomes, collecting PROM data for outcome follow-up or quality measurement is resource-intensive and often yields low response rates [3–6]. A more efficient way of collecting, measuring, and analyzing patient outcomes at scale is needed to allow for continuous system-wide monitoring of surgical success, optimize resource allocation, and facilitate large-scale quality measurement and improvement [7,8].

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Patient interactions with the health care system are documented in the electronic health record (EHR) [9]. Documented clinical encounters offer a wealth of information regarding patient symptoms, postoperative rehabilitation progress, subjective assessment of their recovery, emerging complications, interventions, and other clinical data. Clinicians routinely read and interpret unstructured records in the EHR to gain insight into patient status and the trajectory of outcomes after surgery [10,11]. However, scaling this type of labor-intensive manual review to determine the outcomes of an intervention across a large population is impractical.

Large language models (LLMs) are artificial intelligence systems that are designed to process, interpret, and generate human language [12]. The LLMs have shown potential in performing complex tasks that require human-like reasoning, including generating progress notes, operative reports, and responses to patient questions regarding arthroplasty [12]. Furthermore, LLMs and other natural language processing tools have been deployed for complex clinical outcomes research [13]. Because LLMs are able to process and interpret large volumes of text rapidly, it is conceivable that LLMs may be able to process unstructured notes and other documents in the EHR to infer patient outcomes following an intervention. Developing and widely applying these powerful analytical tools for estimating outcomes after arthroplasty holds tremendous potential for monitoring patient progress, automating large-scale outcome analysis, and driving more personalized, data-driven care.

In that context, the purpose of this study was to (1) analyze the ability of an LLM to interpret unstructured data in the EHR to differentiate patients who achieved minimum clinically important difference (MCID) for the Knee Osteoarthritis Outcome Score–Joint Replacement (KOOS-JR) at 1-year follow-up and (2) determine the ability of a blinded orthopaedic surgeon to achieve the same task, thus providing an estimate of what a properly trained LLM may be able to achieve.

Methods

After obtaining institutional review board approval, a multicenter retrospective case-control pilot study was conducted across three Veterans Affairs (VAs) medical centers in the United States. The study sample was drawn from a cohort of 587 patients who underwent primary TKA between March 19, 2015, and March 9, 2017, and had preoperative and 1-year postoperative KOOS-JR scores [14]. For this study, a total of 80 patients were selected, comprising 40 patients who met the 1-year MCID for improvement of the KOOS-JR and 40 patients who did not. MCID was defined as a 20-point improvement in KOOS-JR scores based on an anchor-based method [14].

We used ChatGPT 3.5 (ChatGPT), a LLM that is not compliant with the Health Insurance Portability and Accountability Act, for this study. Because of this, we needed to anonymize chart information to export it from our health system EHR and have it processed by ChatGPT. Thus, for practical reasons, not all chart information could be entered into ChatGPT. We thus selected what we considered to be high-yield chart entries for interpretation by ChatGPT. For each patient, unstructured EHR data were extracted for two time periods: from zero to six months preoperatively and nine to 15 months postoperatively (the six months before surgery and the six months centered on the 1-year postoperative date). The extracted notes included orthopaedic, primary care, and emergency department notes. These notes were then manually anonymized by removing all protected health information, but the structure and verbiage were retained to ensure fidelity of the text content. Prompts for ChatGPT were written, and the anonymized preoperative and postoperative chart notes were entered for each patient.

ChatGPT was queried as to whether the patient had achieved a meaningful improvement after TKA, with the output recorded and compared to the known KOOS-JR outcome. A new chat was started, and the model reprompted for each patient. The prompts and interaction flow with ChatGPT are depicted in [Figure 1](#).

Simultaneously, a blinded board-eligible orthopaedic surgeon reviewed the EHR for each patient and provided a binary assessment of whether the patient had achieved an improvement at one year after surgery. We did this not to set up a “man versus machine” comparison, but to see if there was any possibility of improving on the performance of the LLM and to identify where additional relevant information may reside in the EHR. The orthopaedic surgeon was able to review all records in the EHR to perform this evaluation. The surgeon also documented specific EHR data points that contributed to this conclusion, and their binary classifications were compared to the known KOOS-JR outcomes.

A correct classification was defined as assigning the patient to the correct outcome (MCID achieved or not). For each classification group (MCID achieved or not), the percent correct (with 95% confidence limits of a proportion) was calculated for ChatGPT and the orthopaedic surgeon. A McNemar’s test was conducted to statistically compare the correct classifications between the two methods in each group, and statistical significance was assessed using *P*-values, with a threshold of *P* < 0.05 to determine significant differences between ChatGPT and the orthopaedic surgeon’s performance in identifying both improvement and nonimprovement cases. In comparative tests, we excluded cases where ChatGPT did not provide a response.

Also, we evaluated the cases where ChatGPT’s responses and the surgeon’s responses differed to understand how one could improve ChatGPT’s automated assessment of outcomes.

Results

ChatGPT provided a “yes” or “no” classification in 78 of 80 cases, while the orthopaedic surgeon provided a classification in all 80 cases. In two cases, ChatGPT was unable to provide a response due to insufficient information.

For the 40 patients who achieved significant improvement (MCID met), ChatGPT correctly identified improvement in 38 cases (95%, confidence interval [CI; 88.25 to 100]), returned “not improved” in one case, and did not provide a response in one case. In comparison, the orthopaedic surgeon correctly identified improvement in 36 cases (90%, CI [80.70 to 99.30]) and incorrectly classified four cases as “not improved.” There was no significant difference in performance between ChatGPT and the orthopaedic surgeon in correctly determining improvement in patients who actually met MCID for the KOOS-JR (*P* = 0.375).

For the 40 patients who did not achieve significant improvement (MCID not met), ChatGPT correctly identified “not improved” in 13 cases (32.5%, CI [17.99 to 47.01]), incorrectly identified improvement in 26 cases, and did not provide a response in one case. In comparison, the orthopaedic surgeon correctly identified “not improved” in 25 cases (62.5%, CI [47.50 to 77.50]) and incorrectly classified 15 cases as “improved.” The orthopaedic surgeon was significantly more likely to correctly identify nonimprovement compared to ChatGPT (*P* = 0.0041).

Confusion matrices for the performance of ChatGPT and the orthopaedic surgeon can be found in [Tables 1 and 2](#), respectively. ChatGPT made more false positive errors, while the orthopaedic surgeon’s errors were more balanced between false positives and false negatives. The sensitivity, specificity, positive predictive value, negative predictive value, and accuracy for both ChatGPT and the orthopaedic surgeon in evaluating postsurgical improvement are presented in [Table 3](#).

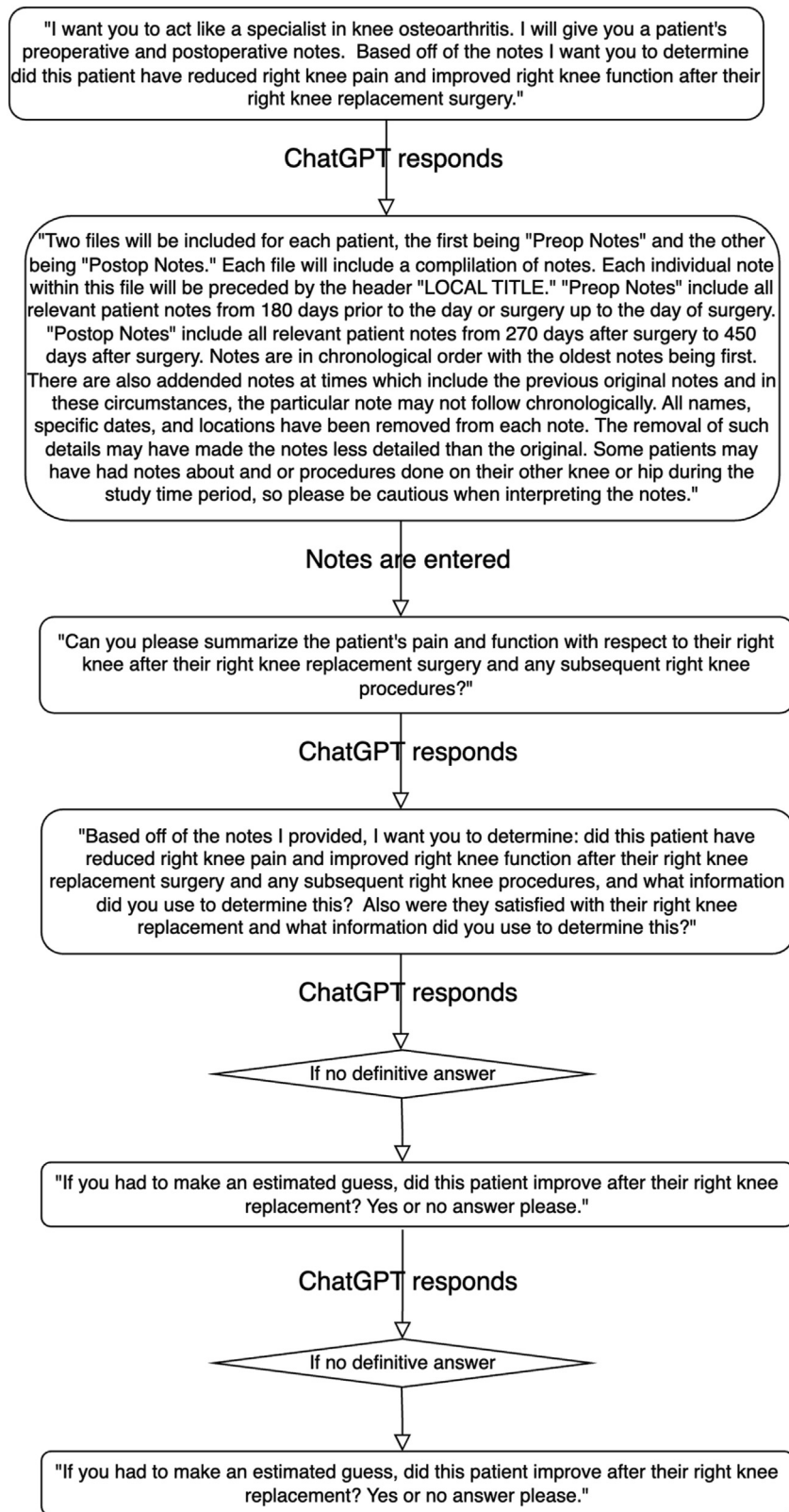


Figure 1. Flow diagram of prompts and interaction with ChatGPT.

We sought to identify patterns in the orthopaedic surgeon's rationale for correctly categorizing 13 patients as nonimproved that the LLM incorrectly categorized as improved. The surgeon's notes revealed several consistent indicators of poor outcome beyond pain

and stiffness (some patients had multiple of these indicators): negative emotional descriptors like patient frustration (three cases), workup for infection or loosening (seven cases), reoperation including manipulation under anesthesia (five cases), a new injury

Table 1
Confusion Matrix of ChatGPT's Classification of Improvement After Total Knee Arthroplasty (TKA).

ChatGPT Outcome Prediction	Actual Improved	Actual Not Improved
ChatGPT predicted improved	38 (TP)	26 (FP)
ChatGPT predicted not improved	1 (FN)	13 (TN)

TP, true positives; TN, true negatives; FP, false positives; and FN, false negatives, are shown.

leading to acute worsening (four cases), and alternative diagnoses (sciatica, overstuffed patella, or overuse syndrome) to explain persistent pain (three cases). While the LLM appeared not to weigh such factors as strongly suggestive of a poor outcome, the surgeon recognized these events as key signals that a patient likely had not achieved clinically meaningful improvement.

Discussion

In this study, we evaluated the ability of a general-purpose, current-generation LLM to “read and interpret” unstructured EHR notes to infer whether a patient achieved MCID for KOOS-JR one year after TKA surgery. Because the LLM we used was outside of the secure EHR environment, information provided to the LLM required manual deidentification. Given this constraint, we elected to manually deidentify orthopaedic, primary care, and emergency department physician notes for the six months before TKA and for the period nine to 15 months after surgery, as we believed that these notes would be most likely to give the information needed to determine MCID achievement. The LLM identified improvement with high sensitivity (95%) but poor specificity (33%).

In contrast, the orthopaedic surgeon had access to the entire EHR for the patient. We did this to see if there was any possibility of improving on the performance of the LLM and to identify where additional relevant information may reside in the EHR. The orthopaedic surgeon had similar performance in identifying positive results (90% sensitivity), but performed better than the LLM in identifying poor results (63% specificity).

The discrepancy in performance between ChatGPT and the orthopaedic surgeon may be attributed to several factors. The orthopaedic surgeon had access to a more comprehensive dataset, including physical therapy notes, laboratory studies, notes at all time points, and other documentation that the LLM did not. Also, postoperative orthopaedic notes, which were the primary input data for ChatGPT, may overstate patient improvement, as surgeons often perceive outcomes more favorably than patients, particularly in unsatisfied patients who have worse outcomes [15–17]. In addition, surgeons often prioritize documenting serious complications, major adverse events, and other issues requiring medical attention or intervention, which can result in notes that present an overly positive view of patient outcomes when such serious issues are not present. Also as noted in our assessment of the discrepancies between the conclusions of ChatGPT and the surgeon, more subtle signs of trouble that are documented in the EHR such as

Table 2
Confusion Matrix of the Adult Reconstruction Surgeon's Classification of Improvement After Total Knee Arthroplasty (TKA).

Surgeon Outcome Prediction	Actual Improved	Actual Not Improved
Surgeon predicted improved	36 (TP)	15 (FP)
Surgeon predicted not improved	4 (FN)	25 (TN)

TP, True positives; TN, true negatives; FP, false positives; and FN, false negatives, are shown.

Table 3
Sensitivity, Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and Accuracy of ChatGPT and the Orthopaedic Surgeon in Assessing for Improvement After Surgery.

Prediction Performance Metric	ChatGPT (Adjusted for Nonresponses)	Orthopaedic Surgeon (Adjusted for ChatGPT's Nonresponses)	P-Value
Sensitivity (%)	97.44	89.74	0.375
Specificity (%)	33.33	64.10	0.0041
PPV (%)	59.38	71.43	–
NPV (%)	92.86	86.21	–
Accuracy (%)	65.38	76.92	0.0495

The two cases where ChatGPT did not reach a conclusion are omitted.

negative emotional descriptors, workup for infection or loosening, reoperation including manipulation under anesthesia, and alternative diagnoses to explain persistent pain, were detected by the surgeon, but not by ChatGPT as a sign of not achieving MCID for KOOS-JR. Going forward, embedding a LLM behind the EHR's fire-wall would make all chart notes available for analysis. Furthermore, it seems likely that refinements to an LLM's prompt or training, such as instructing it to interpret a negative workup for infection as an ongoing concern or sign of nonimprovement rather than a resolved issue, could improve its ability to detect subtle markers of nonimprovement in medical records. Such work will need to be the subject of further study.

Potential Limitations

There were several potential limitations to this study. Because we had to manually deidentify notes to provide them to the LLM, the data we gave to the LLM were limited to orthopaedic, primary care, and emergency medicine notes from a restricted time period. However, this study serves as a proof of concept, and we speculate that access to additional data could improve future model performance. Indeed, the orthopaedic surgeon, who had access to a much broader data set, including physical therapy notes and laboratory values, was able to perform better than the LLM.

In addition, the LLM was not specifically trained to work with medical record data and discern outcomes following surgery. The orthopaedic surgeon was able to pick up on more subtle indicators of patient nonimprovement than the LLM. Also, we made no attempt to optimize the prompt given to the LLM to improve performance. This remains another topic for further study.

Furthermore, we did not have the LLM look at structured data in the EHR. Such information may be incorporated into a predictive model of benefit to improve model performance. Moreover, we did not test intraobserver and interobserver variability in surgeon assessment of outcomes. Assessing the EHR for signs of meeting KOOS-JR MCID has a subjective component, and it is possible that another observer may have come to different conclusions. However, the point of having a surgeon manually assessing the EHR was to see if there was room for improvement in LLM performance, and the fact that the surgeon performed better than the LLM reveals that there is.

Also, the VA health care system is not a closed system, and it is possible for veterans who have poor outcomes to seek care in other health care systems. This could bias the results of an automated EHR-based surgical outcome surveillance system, possibly making the overall outcomes appear better than they really are. This problem is not unique, however, to EHR-based outcome surveillance. Survey response rates can be low, and patients who have lower satisfaction and other patient-reported outcome scores have been found to be significantly less likely to respond to surveys [18]. Furthermore, the cohort consisted of a VA population, which may not be fully generalizable to broader patient populations.

Despite these limitations, we demonstrated that there is potential for an LLM to predict patient outcomes by analyzing large volumes of unstructured clinical data from the EHR. This study provides a practical foundation for future investigations of LLMs in assisting in large-scale data analysis, quality improvement, and outcomes prediction.

Conclusions

In conclusion, this study demonstrated that a currently available, general-purpose LLM reading selected unstructured notes from the EHR had a 97% sensitivity to detect achievement of MCID for KOOS-JR one year after TKA but only a 33% specificity. It is unknown at this point whether this level of performance is sufficient for quality monitoring purposes, but information in the EHR exists to improve this performance. We believe that continued refinement and application of these models to clinical data in the EHR hold promise for large-scale analysis of patient outcomes, providing the opportunity for early identification of poor outcomes and system-wide quality improvement efforts in orthopaedics and medicine in general.

CRediT authorship contribution statement

Leo Zalikha: Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Thomas S. Hong:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Easton A. Small:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Michael Constant:** Writing – review & editing, Formal analysis, Data curation. **Alex H.S. Harris:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Nicholas J. Giori:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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